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OLG Model:

$$\ln c^y + \beta \ln c^o \rightarrow \max$$

$$\left. \begin{array}{l} c^y = w - s \\ c^o = s(1+r) \end{array} \right\} c^y + \frac{c^o}{1+r} = w$$

$$\underbrace{c^y = \frac{1}{1+\beta} w}_{\text{properties of log}}; \underbrace{\frac{c^o}{1+r} = \frac{\beta}{1+\beta} w}_{\text{properties of log}} \Rightarrow c^o = \frac{\beta(1+r)w}{1+\beta}$$

↳ properties of log

$$s = w - c^y = w - \frac{1}{1+\beta} w = \frac{\beta}{1+\beta} w = s$$

$$\text{Firm: } Y = K^\alpha L^{1-\alpha} \Rightarrow f(k) = \frac{K^\alpha L^{1-\alpha}}{L} = k^\alpha$$

$$w = f(k) - f'(k)k = k^\alpha - \alpha k^{\alpha-1} \cdot k = k^\alpha - \alpha k^\alpha = \underline{k^\alpha(1-\alpha)}$$

$$r = f'(k) = \underline{\alpha k^{\alpha-1}}$$

$$\text{MCC: } \hookrightarrow \text{Capital: } s = k(1+n) \Rightarrow \frac{\beta}{1+\beta} w = k(1+n) \Rightarrow \frac{\beta}{1+\beta} k^\alpha(1-\alpha) = k(1+n)$$

$$\Rightarrow k^{1-\alpha} = \frac{\beta}{1+\beta} \frac{(1-\alpha)}{(1+n)} \Rightarrow \underline{k^c = \left(\frac{\beta}{1+\beta} \frac{(1-\alpha)}{(1+n)} \right)^{\frac{1}{1-\alpha}}}$$

MCC:

$$\hookrightarrow \text{Wage: } w = (1-\alpha)k^\alpha \Rightarrow (1-\alpha) \left[\frac{\beta}{1+\beta} \frac{1-\alpha}{1+n} \right]^{\frac{\alpha}{1-\alpha}}$$

$$c^y = \frac{1}{1+\beta} (1-\alpha) \left[\frac{\beta}{1+\beta} \frac{1-\alpha}{1+n} \right]^{\frac{\alpha}{1-\alpha}}$$

$c^o = \dots$

$$r = \alpha k^{\alpha-1} = \alpha \left[\frac{\beta}{1+\beta} \frac{1-\alpha}{1+n} \right]^{\frac{1-\alpha}{\alpha-1}} = \underbrace{\frac{\alpha(1+\beta)(1+n)}{(1-\alpha)\beta}}$$

SPP: $f'(k) = n$

$$\alpha k^{\alpha-1} = n \Rightarrow k^{sp} \left[\frac{\alpha}{n} \right]^{\frac{1}{1-\alpha}}$$

$r < n$: Dynamic Inefficiency

$t=0$: $\gamma \rightarrow 0$ 1 unit of consumption good

$$L_t \cdot 1 = L_{t-1} \cdot X \Rightarrow X = (1+n) \text{ bc pop. growth}$$

$(1+n) > (1+r) \therefore$ better off for c^o

Schemes to Correct D.I.

① Gov't Debt:

$$t: B_{t+1}; \quad \frac{B_{t+1}}{L_t} = b \text{ (constant)}$$

$$\tau_t \text{ (total tax rev)} = \tau_t \cdot L_t$$

↳ Lump-sum tax on young

② Gov't Budget

$$t: B_t(1+r_t) = B_{t+1} + \tau_t$$

per-Capita:

$$b(1+r_t) = \underbrace{\frac{B_{t+1}}{L_t}}_b \cdot \underbrace{\frac{L_t}{L_{t-1}}}_{1+n} + \underbrace{\frac{\tau_t}{L_t}}_{\tau_t} \underbrace{\frac{L_t}{L_{t-1}}}_{1+n}$$

$$b(1+r_t) = b(1+n) + \tau_t(1+n)$$

$$b(r_t - n) = \tau_t(1+n) \Rightarrow \tau_t = \frac{b(r_t - n)}{1+n}$$

Consumer Problem

$$\ln c_t^y + \beta \ln c_{t+1}^o \rightarrow \max_{c_t^y, c_{t+1}^o, s_t}$$

$$\left. \begin{aligned} c_t^y &= w_t - \tau_t - s_t \\ c_{t+1}^o &= s_t(1+r_{t+1}) \end{aligned} \right\} c_t^y + \frac{c_{t+1}^o}{1+r_{t+1}} = w_t - \tau_t$$

$$c_t^y = \frac{1}{1+\beta} (w_t - \tau_t), \quad s_t = w_t - \tau_t - \frac{1}{1+\beta} (w_t - \tau_t) = \frac{\beta}{1+\beta} (w_t - \tau_t)$$

MCC(k):

$$s_t L_t = K_{t+1} + B_{t+1} \Rightarrow s_t = k_{t+1}(1+n) + b$$

$$\text{in SS: } s = k(1+n) + b$$

$$\begin{aligned} \frac{\beta}{1+\beta} (w - \tau) &= \frac{\beta}{1+\beta} \left((1-\alpha)k^\alpha - \frac{b(r-n)}{1+n} \right) \\ &= \frac{\beta}{1+\beta} \left((1-\alpha)k^\alpha - \frac{b(\alpha k^{\alpha-1} - n)}{1+n} \right) = k(1+n) + b \end{aligned}$$

Gov't interferes w/ capital mkt by messing w/ agg. k. choose b s.t. $r=n$

Suppose $r=n$, find b^* via:

$$\frac{\beta}{1+\beta} \left((1-\alpha) (k^{sp})^\alpha \right) = k^{sp} (1+n) - b$$

Gov B.C.:

$$B_t(1+\tilde{r}) = B_{t+1} + \tilde{\gamma} \tilde{L}^0 \Rightarrow \frac{B_{t+1}}{B_t} = 1+n$$

Other Remedy: Social Security
2-type:

- ① fully-funded
 - ② pay as you go
- } Pay lump τ to get P

Pension Balance Eq:

↳ Fully-funded: $P = \tau(1+r)$

↳ pay as go: $L_{t-1}P = L_t \tau \Rightarrow P = (1+n)\tau$

MCC(k):

FF: $S_t L_t + \tau L_t = K_{t+1} \Rightarrow S + \tau = k(1+n)$

PAYG: $S_t L_t = K_t \Rightarrow S = (1+n)k$ (less saving tho)

Consumer Problem:

$$\ln c^Y + \beta \ln c^O \rightarrow \max_{c^Y, c^O, s}$$

$$\left. \begin{array}{l} c^Y = W - \tau - S \\ c^O = S(1+r) + P \end{array} \right\} c^Y + \frac{c^O}{1+r} = \underbrace{W - \tau + \frac{P}{1+r}}_{\text{Lifetime Wealth}}$$

$$c^Y = \frac{1}{1+\beta} \cdot \text{LTW}; \quad c^O = \frac{\beta}{1+\beta} \cdot \text{LTW}(1+r)$$

$$\begin{aligned} S = W - \tau - c^Y &= W - \tau - \frac{1}{1+\beta} \left(W - \tau + \frac{P}{1+r} \right) \\ &= \frac{\beta}{1+\beta} (W - \tau) - \frac{1}{1+\beta} \left(\frac{P}{1+r} \right) \end{aligned}$$

PAYG:

$$S = k(1+n)$$

$$\frac{\beta}{1+\beta} (W - \tau) - \frac{1}{1+\beta} \left(\frac{P}{1+r} \right) = k(1+n)$$

$$\frac{\beta}{1+\beta} \left((1-\alpha)k^\alpha - \tau \right) - \frac{1}{1+\beta} \left(\frac{r(1+n)}{1+r} \right) = k(1+n)$$

set $r=n$; $k=k^{sp}$ (not always case. desirable here)

$$\frac{\beta}{1+\beta} \left((1-\alpha)k^{sp\alpha} - \tau \right) - \frac{1}{1+\beta} (\tau) = \left[\frac{\alpha}{n} \right]^{\frac{1}{1-\alpha}} (1+n)$$

$\Rightarrow$ solve for τ .